# Chip Convergence

## Verification-first agents for chip design

**NVIDIA** RTL PPA · **NXP** SoC generation · **ASU** DRC repair

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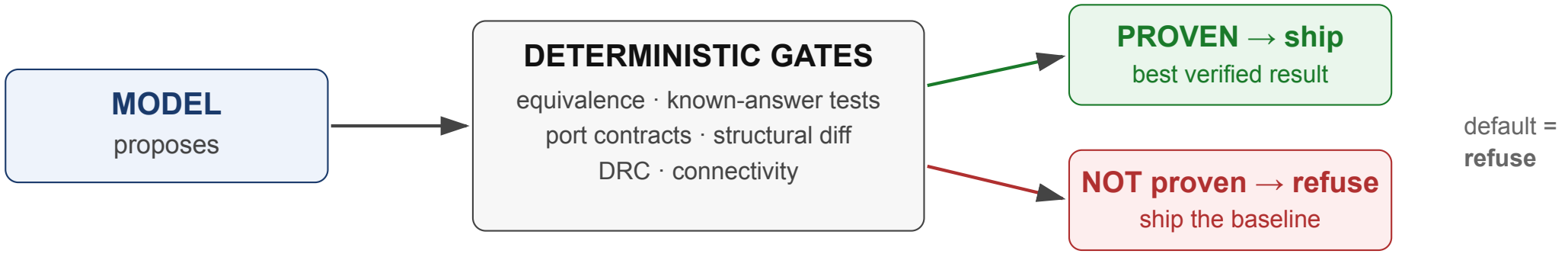
Harikrishnan KC (solo) · greatharikrishnan@gmail.com

# 1 · One thesis, three problems

A language model will hand you hardware that is **smaller, faster — and wrong**.
Plausible-but-wrong RTL is worse than no RTL: it passes review and fails in silicon.

So we built the **verifier first** and put the model behind it.




The model is a **search heuristic over a verified space** — never an authority.
**The system's default is to ship nothing.**

# 2 · NVIDIA — improve PPA, stay functionally identical

**Given:** 7 IPs (async_fifo, sha512, NVDLA, OpenTitan aes/ascon/kmac/prim), their testbenches, a Yosys + OpenSTA ASAP7 flow.
**Scored on:** area × delay vs baseline (**ADP**, lower is better), gated on correctness.

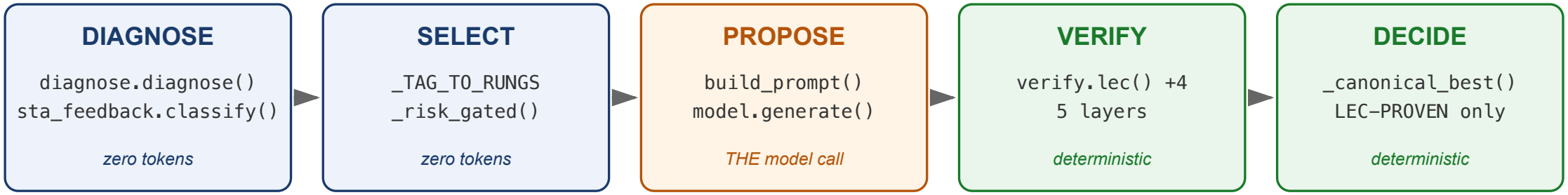
**Five verification layers** — every candidate, every round:

| # | layer | what it answers |
|---|---|---|
| 1 | lint | structurally sane? |
| 2 | compile | does it elaborate? |
| 3 | **gate** | the IP's own testbench — fail-closed: nonzero rc = FAIL |
| 4 | **LEC** | yosys *proves* equivalence to pristine |
| 5 | dualsim | cycle-exact co-simulation, baseline vs candidate |

**PROVEN binds all of:** `rc==0` + success line + `total>0` + `proven==total` + `unproven==0`.
Anything else is INCONCLUSIVE — never a pass.

**Canonical = LEC-PROVEN.** An unproven candidate is never shipped, however good its ADP.

# 3 · NVIDIA — the flow, end to end




STA tells us WHICH file and WHAT structure — so the model is asked for one named transform on one named cone, never “make it faster”.

**Green is free. Orange is the only probabilistic step. The last two stages are where the model gets no vote** — five verification layers, then a selector that admits only formally-proven candidates and otherwise ships the baseline untouched.

# 4 · NVIDIA — results, and the ones we refused

| IP | ADP | assurance |
|---|---|---|
| **prim** | **0.5824** | LEC PROVEN + dualsim · power −70% |
| **sha512** | **0.7266** | full 5-layer · timing −97 → **+335 ps (MET)** |
| ascon | 0.97918 | LEC PROVEN + dualsim |
| aes | 1.0001 | differential-only — honestly *"explored-no-adp-gain"* |
| kmac | 1.0 | bounded negative — headroom is inside the masked core |
| **async_fifo** | **1.0 (baseline)** | **deliberately reverted — next slide** |

**Two refusals worth more than the wins:**

- **ascon:** a candidate *beat* the banked ADP — `lec_diagnostic` found a real counterexample → **refused**. The gate is load-bearing, not decorative.
- **sha512, live:** a candidate at **ADP 0.6546** — far better than anything we ship — was **refused for not being LEC-PROVEN**. The system gave up a 35 % headline number rather than ship something it could not prove.

# 5 · NVIDIA — the 4 % win that was the bug

Our agent found a **4 % ADP win** on an asynchronous FIFO.
It passed lint. Compile. The testbench. Formal equivalence **PROVED** it. Cycle-exact differential simulation passed. **Five for five** — our best-verified candidate.

## It was broken.

It removed the registers on the **Gray-code pointer** and computed `gray(bin)` combinationally. The logic function is identical — *which is why formal proved it*. But at a **clock-domain crossing** a combinational encoder **glitches** mid-transition, and the receiving domain samples asynchronously. Intermittent pointer corruption.

We reverted to the registered baseline, then rebuilt the *separable* part of the optimization on the safe base and measured it: **ADP 0.9984 — noise.**

> **The entire 4 % "win" *was* the hazard.**

# 6 · NVIDIA — NVDLA: whole-design formal at scale

**~950,000 cells.** Whole-design equivalence **PROVEN: 381,209 / 381,209 `$equiv` cells** through the production recipe — uncommon at this scale.

**Release packet: 6/6** — every gate control demonstrated live, including the negatives:

| control | result |
|---|---|
| pristine gate must PASS | ✅ 1 passed / 0 failed |
| **mutant gate must FAIL** | ✅ 0 passed / 1 failed |
| LEC positive PROVEN · LEC negative not-proven | ✅ ✅ |
| candidate-survival tripwire · determinism ×2 | ✅ ✅ |

**We found our own tooling lying.** The packet sat at 5/6 for a day. The failing check was a bug in *our log parser* — it counted the word "failed" inside the runner's help banner (`exceeded => failed`) as a test failure. Fixed, re-run, 6/6 — **and the negative control still fires**, proving the fix did not simply blind the gate.

**Not claimed:** an NVDLA PPA result. The campaign plumbing is unfinished; on a hidden NVDLA-like case the agent attempts, cannot prove, and **ships baseline**.

# 7 · NXP — generate a secure SoC from a block diagram

**Given:** an architecture document and a testbench port skeleton. **Produce synthesizable RTL.**

Model reads the diagram → emits **YAML specs per IP** → `rtl_gen_lib` generates Verilog → model stitches the top → **correctness firewall** → gate.

**The firewall — deterministic, non-bypassable, ≤3 error-fed re-prompts:**

- **YAML validator** — typed errors *before* any RTL is generated
- **Port contract** — token-level diff of the top's ports vs the skeleton's DUT instantiation
- **Reset lint** — raw POR may feed *only* the synchronizer; one common synchronized net
- **Structural diff** — instance census, IRQ reachability, bus connectivity
- **Port-direction gate** — an instance's output may never be driven by a constant

**Result (easy tier): 2 model calls, ~42 s** — GATE **30/30**, KAT **79/79** against a golden oracle *and* **79/79** against the model's own oracle, STG differential cycle-identical over **3,662** cycles.

Two oracles on purpose: self-consistency proves the model agrees with itself; the golden oracle proves it is right.

# 8 · NXP — the night the hidden problems landed

The organizers released **medium** (2×3 TileLink NoC AES SoC) and **hard** (multi-domain crypto SoC: 4×3 NoC, 4 AES, 2 DMA) — architectures the agent had never seen.

Running them exposed **nine defects in our own agent.** The worst three:

| defect | effect |
|---|---|
| top-module name hardcoded to the easy problem | every other tier failed elaboration |
| prompt budget sized for 8 IPs | **14 of 22** IP interfaces never shown → the model *guessed* ports |
| `top_ports` parsed `#(parameter…)` as the port list | **any parameterised module reported zero ports** — silently disabling direction checks |

### After the fixes — verified with `iverilog` outside the agent:

| tier | top derived | contract | elaborates |
|---|---|---|---|
| easy | `secure_periph_soc` | clean, attempt 1 | ✅ 30/30 + KAT 79/79 ×2 |
| **medium** | `noc_aes_soc` | clean, attempt 3 | ✅ |
| **hard** | `crypto_soc` | clean, attempt 2 | ✅ |

Easy is **byte-for-byte unchanged** — every generalization was made tier-aware so the proven

# 9 · NXP — the flow, end to end


No reference design exists — so the proof is CONTRACTS and INDEPENDENT ORACLES, not equivalence. Bad specs die before any RTL is generated.

**Two model calls total.** Everything around them is deterministic: the spec validator runs *before* any Verilog exists, and the stitched top must clear four independent gates — port contract, reset lint, structural diff, port directions — with typed errors feeding up to three repair attempts.

# 10 · NXP — instructions are probabilistic. Gates are not.

A three-step experiment, run last night:

**1. The model already had the information.**
The port directions were *in the prompt*: `output wire [1:0] p0_d_param`.
It tied constants to those outputs anyway — illegal Verilog, elaboration dies.

**2. We added an explicit prohibition.**
*"An OUTPUT port may NEVER be connected to a constant."*
It complied on **medium** (2 runs of 2). It **violated it on hard.**

**3. We added a deterministic gate.**
`instance_port_directions` compares every connection against the real port direction.
Caught **4/4** violations. The typed error feeds the existing repair loop — and medium and hard then reached contract-clean at attempts 3 and 2.

> **The model repaired its own output once the gate told it exactly what was wrong.**

Plus a mechanical last resort — `.m_rready(1'b1)` → `.m_rready()` — so a stubborn violation degrades to an **elaborating** design instead of a zero.

# 11 · NXP — what we do and do not claim

**Claimed, and reproducible:**

- easy tier: 2 calls, 30/30 gate, KAT 79/79 on both oracles
- medium and hard: contract-conformant RTL that **elaborates against the organizer's skeleton**, on architectures the agent had never seen
- stdlib-only agent; a `_yaml_compat` shim that engages only when PyYAML is absent, parses all 8 reference specs byte-identically, and returns an **inert string** for `!!python/object`

**Not claimed — and this matters:**

- **No golden testbench ships for *any* tier**, easy included. So medium and hard cannot be *scored*. "Elaborates and conforms" ≠ "functionally correct".
- The official runner still hardcodes `PROBLEMS = ["easy"]`; we drove the new tiers through the same `Paths.from_info` contract the runner uses.
- Our easy result is **"perfect against our verification stack"** — the organizer's hidden testbench is not in the public checkout.

# 12 · ASU — repair DRC violations in a layout block

**Given:** a KLayout `pya` layout script, its DRC report, connectivity and design rules.
**Scored on:** final-violation-rate (**FVR**, lower is better), gated on the repair being valid and connectivity-preserving.

## The result that shows judgment: we took the model out.

The winning repair is **deterministic** — replace each flagged multi-cut via array with one continuous via **bar**, derived directly from the rule. No model call at runtime.

**Why that was forced, and why it was right:** the official agent image is `python:3.10-slim` with **no KLayout binary**, run `--read-only`. Our development agent measured every candidate with the real evaluator to keep-best — impossible there.

So the submission agent applies the proven transform and emits an artifact **byte-identical** to the independently re-scored development output.

We used the model to *find* the transform. We did not need it to *apply* the transform.

# 13 · ASU — v2 Rev3, all seven blocks, electrically proven

Same-day arc: a 3-round independent layer-aware review found the v1 via-bar created **49 electrical merges invisible to the published checker**; we rebuilt with per-side electrical guards + a new **track-shift** pass, re-proved every block, and resubmitted (official runner rehearsal, zero model calls).

| block | kind | valid | conn | v1 FVR | Rev3 FVR | electrical partition |
|---|---|---|---|---|---|---|
| Block1 | public | ✅ | ✅ | 0.7295 | **0.5820** | ✅ equal |
| Block2 | public | ✅ | ✅ | 0.7647 | **0.5147** | ✅ |
| Block3 | public | ✅ | ✅ | 0.7640 | **0.3933** | ✅ |
| Block6 | public | ✅ | ✅ | 0.6761 | **0.4130** | ✅ |
| Block7 | public | ✅ | ✅ | 0.6824 | **0.5804** | ✅ |
| **Block4** | **released Jul 25** | ✅ | ✅ | 0.6939 | **0.3741** | ✅ |
| **Block5** | **released Jul 25** | ✅ | ✅ | 0.7941 | **0.4853** | ✅ |

**7/7 eligible** · mean **0.477** (v1: 0.729) · **Block4/5 blind = its two best** · no-op **≈1.25–1.32**

# 14 · ASU — why it generalized

No tuning. No retraining. No code change. The hidden blocks were released and the agent scored them in the same band as the public ones.

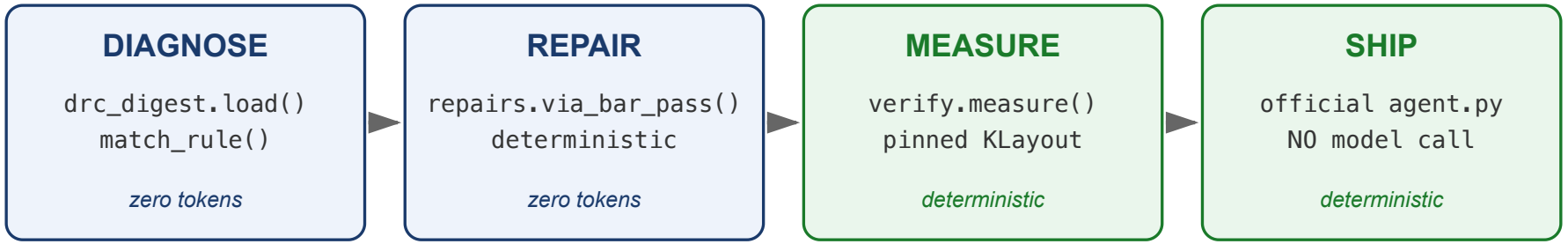
**Because the repair is derived from the rule, not fitted to the data.**

A model that had been *optimized against the five public blocks* would have no reason to transfer. A transform decoded from the DRC rule itself has every reason to.

**Three honesty points we put on the slide before anyone asks:**

- **No-op is not FVR 1.0.** FVR is the evaluator's *fresh* DRC total ÷ the *supplied reference* total, and those are not count-identical: public no-op FVR is **≈1.25–1.32**.
- **Our own review disqualified our own best number.** The withdrawn v2 draft scored FVR 0.24–0.39 — but inherited 49 electrical shorts the published checker cannot see. Rev3 gives back part of the DRC win to make the connectivity claim *provable* (immutable-anchor partition equality, per block, in the repo).
- **Fail-closed by construction.** Landings/moves that cannot be proven electrically safe keep their original geometry; on clean or unfamiliar layouts every pass is a verified no-op.

# 15 · ASU — the flow, and why the model left it




The model helped us FIND the transform. The shipped agent does not call it — the official image has no KLayout, so measure-and-keep-best is impossible there.

**Two agents.** The development agent measures every candidate with the organizers' own evaluator and keeps the best — that is how we *found* the via-bar transform. The shipped agent is seven functions, stdlib-only, and makes zero model calls.

# 16 · Three domains, one finding

| | the model was told | what happened | what worked |
|---|---|---|---|
| **NVIDIA** | — | a candidate passed **all five layers incl. LEC-PROVEN** and was still unsafe (CDC glitch) | a **structural invariant**, not more checking |
| **NXP** | port directions, in the prompt | ignored them; obeyed an explicit rule on one problem, broke the other | a **deterministic gate** — 4/4, every time |
| **ASU** | — | hidden blocks scored in-band, untouched | the transform is **derived from the rule**, not fitted |

## Deterministic gates beat model instructions.

That is what makes **unseen inputs** survivable — and unseen inputs are the whole game.

# 17 · What formal and simulation structurally cannot see

LEC and zero-delay simulation reason in a model where **glitches and physical side-channels do not exist.** Two classes of edit therefore pass every functional check and are still broken:

**1 · Side-channel masking removal** — aes S-box, kmac DOM-masked Keccak.
Un-masking preserves the logic function *exactly*, so **LEC returns PROVEN** while the countermeasure is destroyed. Formally equivalent, cryptographically broken.

**2 · CDC glitch hazards** — async_fifo (slide 4).
Neither formal nor zero-delay simulation can represent a glitch.

## You cannot fix this by adding more checking.

The bug lives **outside the model the checkers use**. The answer is **structural policy**: per-IP forbidden edit zones (FENCEs — implemented, mandatory for NVDLA), and a registered-crossing invariant for CDC.

Honest status: the FENCEs are enforced in tooling; the CDC invariant was implemented this week after we caught the agent re-selecting the candidate we had manually reverted — by running our own submission the way a judge would.

# 18 · Every claim, one command

| track | headline | verify it |
|---|---|---|
| **NVIDIA** | prim **0.5824**, sha512 **0.7266**, both LEC-PROVEN | `submission/<ip>/manifest.json` — per-layer assurance is recorded per artifact |
| | NVDLA formal **381,209/381,209**; packet **6/6** | `python3.12 -m ppa.release_control --ip nvdla` |
| **NXP** | easy 2 calls · **30/30** · KAT **79/79 ×2** | `python3 nxp_agent.py --model stub` |
| | medium + hard **elaborate**, unseen | `iverilog -g2005 <rtl>/*.v <tb_top_skeleton.v>` |
| **ASU** | **7/7** eligible, Rev3 mean FVR **0.477**, electrical partition proven | `asu_v2/tests/run_controls.sh` + `evaluator/evaluate_repair.py --case Block4` |

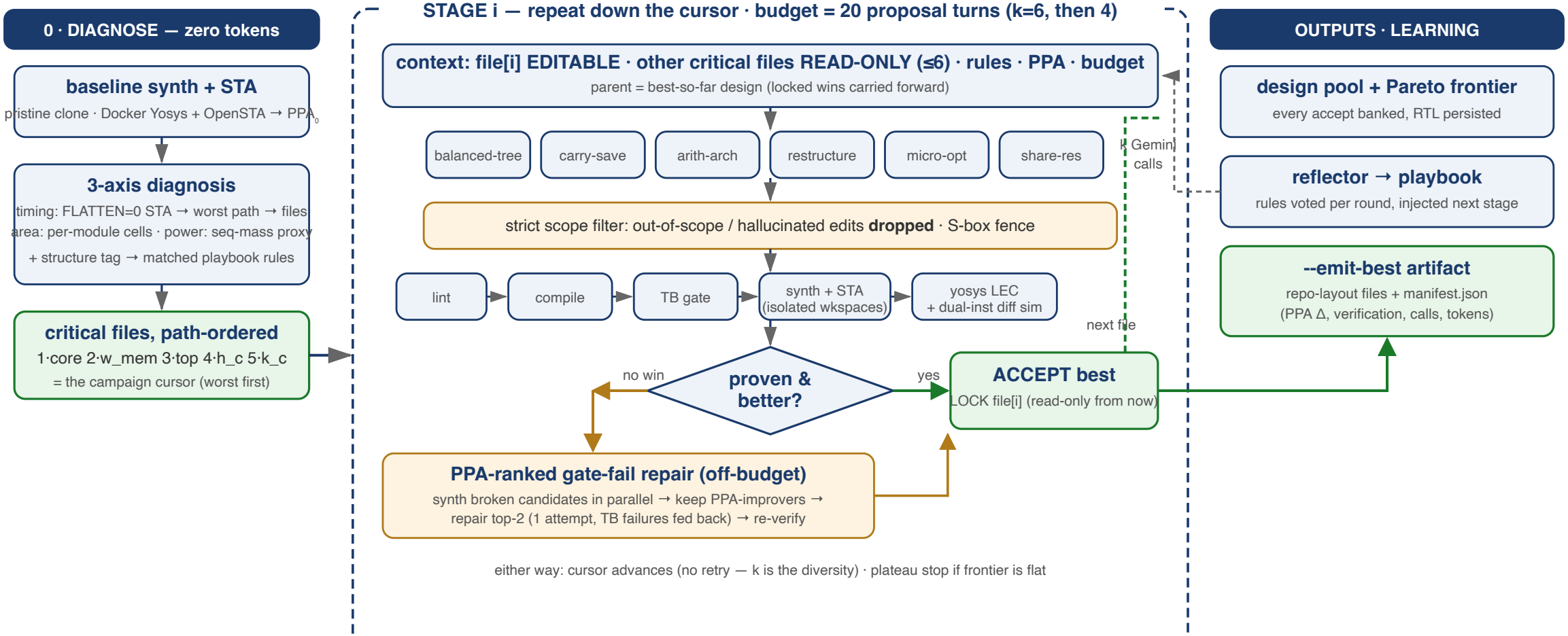
**Reproducibility is the point.** NXP and ASU reproduce byte-identically. The NVIDIA *artifacts* re-synthesize exactly; re-running the *search* is stochastic — and we say so.

**The architecture, in one line:** the model proposes, deterministic gates dispose, and the default is to ship the baseline.

# Backup

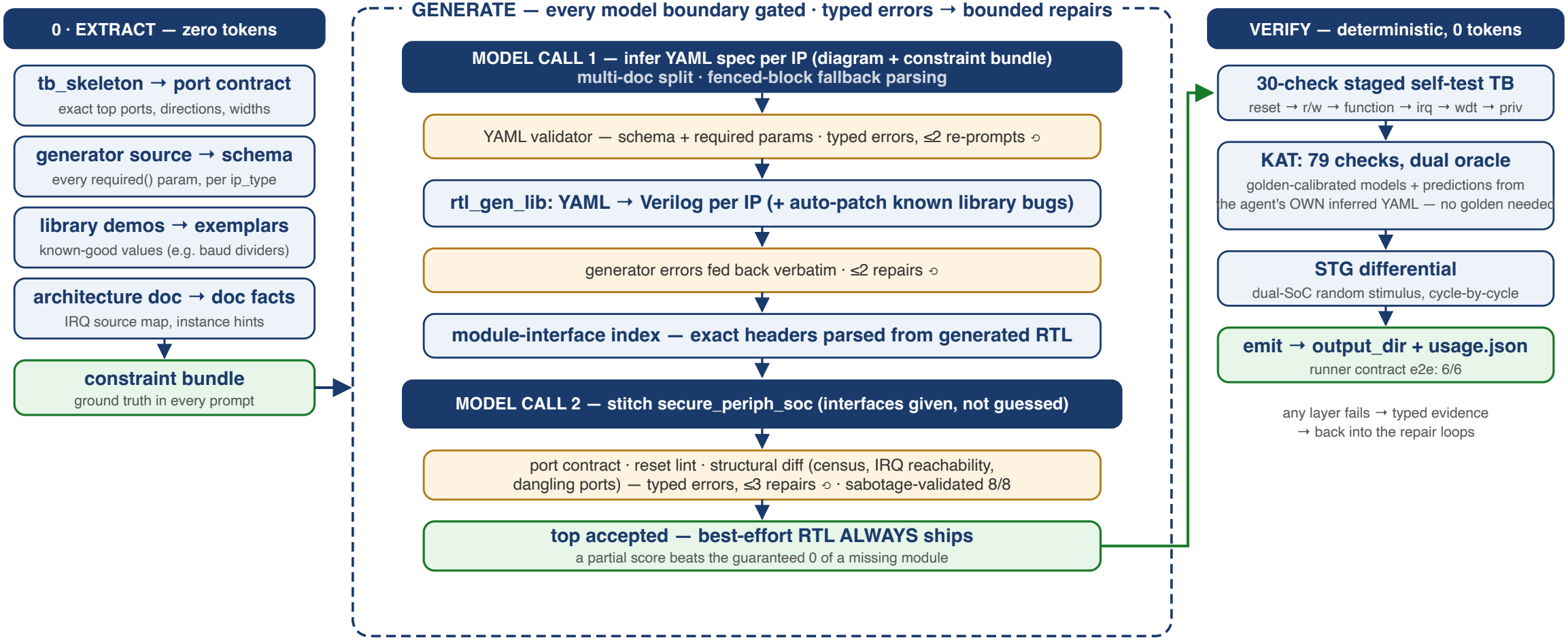
Flow diagrams · number sheet · what we do not claim

# B1 · NVIDIA — the full flow



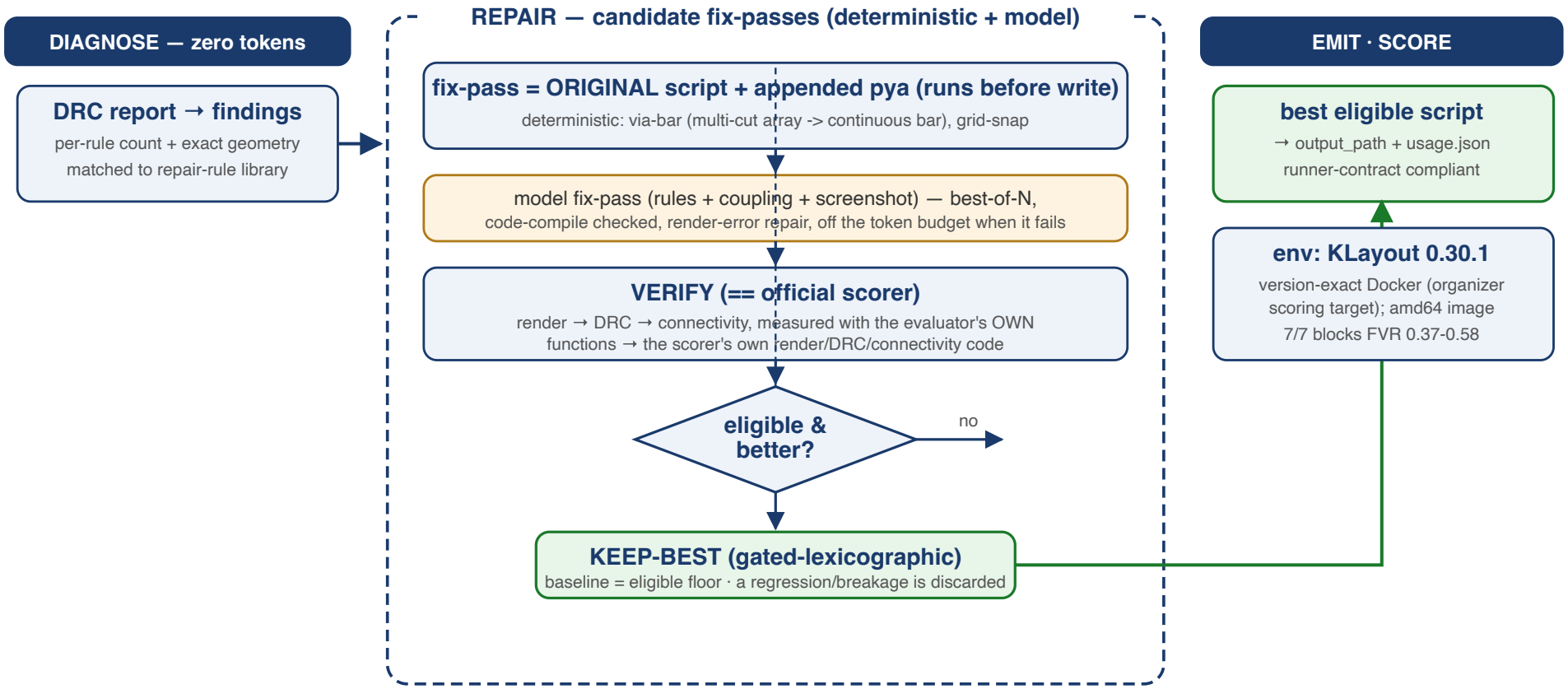

Banked: sha512 **ADP 0.7266** full 5-layer · prim **ADP 0.5824** LEC-PROVEN + dualsim (power −70%) · async_fifo ships **baseline** by design.

# B2 · NXP — the full flow




Live: **2 model calls, 42 s** → 30/30 · KAT 79/79 (both oracles) · STG 3662 cycles, **0 differing** vs hand-built reference.

# B3 · ASU — the full flow




Same verification-first spine as NVIDIA/NXP: diagnose (0 tokens) → propose → verify == scorer → keep-best (electrically gated) → emit. **v2 Rev3: FVR 0.37–0.58 on all 7 blocks.**

# B4 · Verified number sheet

**NVIDIA** (ADP, baseline = 1.0, lower better)
prim **0.5824** LEC-PROVEN+dualsim, power −70% · sha512 **0.7266** full 5-layer, WNS −97 → +335 ps
ascon **0.97918** PROVEN · aes **1.0001** differential-only · kmac **1.0** bounded negative
async_fifo **1.0** deliberate revert · NVDLA formal **381,209/381,209**, release packet **6/6**
Refused: ascon INEQUIVALENT counterexample; sha512 **0.6546** unproven → refused

**NXP** easy: **2** calls / ~42 s · GATE **30/30** · KAT **79/79** golden + **79/79** model ·
STG differential **3,662** cycles · medium (`noc_aes_soc`) and hard (`crypto_soc`) elaborate,
contract-clean at attempts 3 and 2

**ASU v2 Rev3** 7/7 eligible · FVR **0.374–0.582** (mean **0.477**; v1 was 0.729) ·
electrical partition preserved on all 7 (layer-aware proof) · Block4/5 scored blind: **0.3741 / 0.4853**

**Reproduction, 2026-07-24/25:** fresh GitHub clone → NXP stub 30/30 + KAT 79/79×2; ASU output **byte-identical** (`fdae65dd…`) under `python:3.10-slim --read-only`; NVIDIA prim re-run
improved to **0.5762** LEC-PROVEN.

# B5 · What we deliberately do NOT claim

- **Not claimed:** an NVDLA optimization result. Whole-design formal proves; the campaign plumbing is unfinished and the contract stays `PENDING` — every real model campaign is refused.
- **Not claimed:** an official NXP hidden-testbench score. **No golden testbench ships for any tier.** medium/hard "elaborate and conform" ≠ "functionally correct".
- **Not claimed:** ASU "can never be worse" — it was a net win on all five public blocks; some individual violation classes appear while the total falls.
- **Not claimed:** full 5-layer assurance on prim or aes — their manifests record `gate: SKIP-preexisting` and label the result **equivalence+differential**.
- **Not claimed:** "no model can improve kmac" — bounded probes found nothing; that is a bounded negative, not a universal.
- **Not claimed:** run-to-run identity of the NVIDIA search. A shallower run legitimately produced sha512 0.7814 where a deeper one produced 0.7266. The *artifact* reproduces; the *search* is stochastic.

**Every one of these was, at some point, stated too strongly in our own documents — and corrected.** We audit our own claims.